

Mathematical Foundations of Spline Geometry Analysis

This document provides the theoretical justification for why we use the geometric metric G_e (based on Deviation-From-Linearity and Curvature) to interpret KAN edge computations, and under what conditions it isolates biological signals.

Proposition 1: Reference-Induced Deformation Decomposition

Let the empirical computation along an edge (i, j) be defined as:

$$\Phi_{ij}(x) = W_{ij}^{base} \text{SiLU}(x) + \sum_k w_{ij,k} B_k(x) \quad (1)$$

When evaluating the non-linearity of this function, we compute a geometric test statistic $G(f, X)$ (e.g., integrated squared second derivative or deviation from linear fit) over a defined input distribution X .

The Decomposition Hypothesis:

We assert that the observed geometry can be linearly decomposed into two components:

$$G(f, X) = S(f) + D(X, f) \quad (2)$$

Where:

1. $S(f)$ is the **intrinsic structural component**. This represents the true, task-associated mathematical transformation (e.g., thresholding, saturation) required to solve the biological objective.
2. $D(X, f)$ is the **reference/input-dependent deformation**. This is the artifactual geometry induced by how the function f interpolates the specific input density X , constrained by the flexibility of the B-splines.

The Role of Null Calibration

When evaluating $G(f, X)$ across two different cohorts (e.g., HCP $\rightarrow$ CHCP), a naive evaluation yields $G(f_{real}, X_{CHCP})$. If this value changes, it is mathematically impossible to distinguish whether the true task mechanism $S(f)$ changed, or merely the input density X shifted, altering $D(X, f)$.

To resolve this identifiability failure, we train an empirical null model f_{null} on label-shuffled data. Because Y_{null} contains no task information, the null model requires no intrinsic biological mechanism:

$$S(f_{null}) \approx 0 \quad (3)$$

However, f_{null} is trained on the exact same input distribution X , utilizing the identical spline flexibility and regularization dynamics. Thus, it fully absorbs the input-dependent deformation:

$$G(f_{null}, X) = D(X, f_{null}) \quad (4)$$

By constructing the **Standardized Separation** estimator (or simply evaluating the difference):

$$\Psi = G(f_{real}, X) - G(f_{null}, X) \quad (5)$$

$$\Psi = [S(f_{real}) + D(X, f_{real})] - D(X, f_{null}) \quad (6)$$

Crucial Theoretical Guarantee:

Assuming the baseline deformation is statistically independent of the true signal mechanism (i.e., $D(X, f_{real}) \approx D(X, f_{null})$ in expectation), the calibration yields:

$$\Psi \approx S(f_{real}) \quad (7)$$

This proves that the calibrated metric isolates the intrinsic structural behavior relative to the null expectation, removing the confounding effects of X . If $\Psi > 0$, the edge exhibits non-linearity exceeding what is forced by the mere routing of the input distribution.